Md Siam Hossain Sarker

Software Engineer · Backend Architecture & Scalable Systems

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SUMMARY

Software Engineer and M.Sc. Student in Automotive Software Engineering at **TU Chemnitz** with 3.5 years shipping multi-project backend services, RESTful APIs, and asynchronous pipelines across mining, healthcare, e-commerce, and SaaS. Former Software Engineer at **WebAlive** (Melbourne), where I architected **Hykmah** — a multi-product SSO authentication platform. **IEEE** member, **Springer & IEEE** published researcher, international hackathon finalist.

EXPERIENCE

Software Engineer

Jan 2026 — Sep 2026

WebAlive · Melbourne, Australia (Remote)

- Architected **Hykmah** — unified SSO platform with custom OIDC/PKCE flows in Keycloak and token enrichment across interconnected enterprise SaaS products.
- Built centralised user provisioning service (FastAPI + MongoDB) for cross-product identity lifecycle and fine-grained access control.
- Engineered graduation event and ticketing workflows for **Reed Events & GradPak** across CakePHP and Laravel, powering commencement logistics for Australian universities.
- Delivered backend systems in Grails and C# (GDPR-compliant); conducted cross-repository code reviews, technical guidance, and direct feature demos to international clients in Australia and Canada.

`Java` `Keycloak` `OIDC/PKCE` `FastAPI` `Laravel` `CakePHP` `MongoDB` `Grails` `C#` `RabbitMQ`

Software Engineer

May 2023 — Dec 2025

Bit Mascot (Pvt.) Ltd. · Dhaka, Bangladesh

- Led backend engineering for **Mining Bull** (later evolved into **Mining Hub**), building complex PostGIS geospatial spatial queries, spatial indexing, and high-throughput commodity tracking APIs.
- Juggled multiple client projects simultaneously under delivery pressure (**Mining Hub**, **Nintendo Repair**, **Locum West**), rotating between active sprint priorities and critical production releases.
- Engineered resilient asynchronous background workers (RabbitMQ, Laravel queues) and load-handling strategies to process high-volume query tasks without blocking write paths.
- Integrated third-party APIs including Autobill SDK for Nintendo warranty/repairs billing and SendGrid automation for healthcare locum staff scheduling.

`PHP` `Laravel` `CakePHP` `Node.js` `PostgreSQL/GIS` `MySQL` `RabbitMQ` `Docker` `WebSockets`

Undergraduate Teaching Assistant

Nov 2021 — May 2023

United International University · Dhaka, Bangladesh

- TA for Structured Programming, OOP, DBMS, and Computer Networks labs; one-on-one mentoring in C/C++, Java, PHP, and MySQL.
- **President — UIU App Forum** (2020–2023): led 800+ member developer community with industry-partnered events on web3, mobile, and career development.

SELECTED PROJECTS

Hykmah Ecosystem `Keycloak · FastAPI · MongoDB · Grails · C#` 2026

Unified SSO platform with custom OIDC/PKCE flows; single login across independent products with per-product authorisation. Centralised provisioning service manages cross-product access lifecycle.

Log Archiver `FastAPI · DuckDB · Apache Parquet · RabbitMQ` 2025

Enterprise log archival platform with columnar Parquet storage for efficient analytics queries. RabbitMQ-decoupled ingestion sustains throughput under high load without blocking the write path.

Mining Hub & Mining Bull `Laravel · Node.js · PostgreSQL/GIS` ↗ mininghub.com

REST APIs and GIS spatial query systems for two Australian mining platforms; real-time commodity tracking with geospatial filtering and spatial indexes.

Reed Graduation `CakePHP · MySQL · RabbitMQ` ↗ reedevents.com.au

Graduation service platform for universities with CMS for editable certificate templates. Asynchronous certificate generation via RabbitMQ.

Car-Story `Solidity · Ethereum · React` ↗ github

Decentralised vehicle history on Ethereum smart contracts; removes the need for a trusted central authority. Undergraduate thesis — published at **IEEE TENSYMP 2024**.

EDUCATION

MSc in Automotive Software Engineering 2026 — Present

Technische Universität Chemnitz · Chemnitz, Germany

Faculty of Computer Science · Focus on Automotive Platforms & Software Architecture

BSc in Computer Science & Engineering 2019 — 2023

United International University · Dhaka, Bangladesh

CGPA **3.90 / 4.00** (Summa Cum Laude, German equiv. **1.15**) · 137 credits · 11 academic scholarships · Thesis published at IEEE TENSYMP 2024

SKILLS

LANGUAGES
Java, Python, TypeScript / JavaScript, PHP, C#, Solidity, C/C++

BACKEND
Laravel, FastAPI, Node.js (Express, NestJS), Grails, CakePHP

IDENTITY & SECURITY
Keycloak, OIDC, PKCE, OAuth2, token enrichment, access control

BLOCKCHAIN
Solidity, Ethereum, smart contract architecture

DATABASES
PostgreSQL / GIS, MySQL, MongoDB, DuckDB, MariaDB

ASYNC & MESSAGING
RabbitMQ, WebSockets, Apache Parquet, event-driven pipelines

FRONTEND
React, React Native, jQuery, Ajax, ES6

TOOLING
Docker, Git, Linux, Postman, Selenium, LaTeX, Jira

PUBLICATIONS

Car-Story: A Decentralised Vehicle Information Storage System Based on Blockchain
IEEE TENSYMP 2024 — Ethereum smart contract architecture for trustless vehicle history verification.

UAVs in Green Health Care for Energy Efficiency and Real-Time Data Transmission
Springer LNNS Vol. 665 · ICISS 2023 — UAV-based healthcare data delivery with MobileNet real-time detection.

HONOURS & ACHIEVEMENTS

International Blockchain Olympiad — Award of Distinction 2023
Top 20 nationally, international finalist · 6th IBCOL, Amsterdam · Team "Unchained".

ICPC Asia Dhaka Preliminary — Honorable Mention (×2) 2020 – 2023
Three appearances across teams Songshoptok, CodeSpeed, and Brute Force.

Hult Prize 2022 — 1st Runner-up (UIU) 2022
Competed at the Islamabad Regional Summit.

CSE Project Show — Runner-up 2021 – 2022
System Analysis & Design Lab (Fall '21), Software Engineering Lab (Spring '22).

CERTIFICATIONS

Python for Everybody Specialization — Coursera / University of Michigan (Jul 2020)
PCAP: Programming Essentials in Python — Cisco Networking Academy (Jul 2020)
Introduction to Cloud Development — edX · HTML, CSS, JavaScript (Apr 2022)
Responsive Web Design — freeCodeCamp (Feb 2022)

LANGUAGES & MEMBERSHIPS

Languages: Bengali (native) · English — Professional proficiency (IELTS 6.0 / B2)
IEEE Student Member · **EMK Center** General Member (since 2018)